CARESYNC: AN AI-ASSISTED CLINICAL TRIAGE, TELEMEDICINE, AND DOCTOR RECOMMENDATION ARCHITECTURE FOR DIGITAL HEALTHCARE

A Project Report
Submitted
In Partial Fulfillment of the Requirements
For the Degree of

Bachelor of Technology (B.Tech)
in
Computer Science Engineering (Data Science)

by

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NOVEMBER 2026

DECLARATION

We hereby declare that the project work presented in this report entitled "CareSync: An AI-Assisted Clinical Triage, Telemedicine, and Doctor Recommendation Architecture for Digital Healthcare", in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering (Data Science), submitted to Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow is based on our own work carried out at the Department of Computer Science & Engineering (Data Science), G.L. Bajaj Institute of Technology & Management, Greater Noida.

The work contained in the report is true and original to the best of our knowledge and project work reported in this report has not been submitted by us for award of any other degree or diploma.

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CERTIFICATE

This is to certify that the Project report entitled "CareSync: An AI-Assisted Clinical Triage, Telemedicine, and Doctor Recommendation Architecture for Digital Healthcare" done by Tushar Jeswani (Roll No. 2410912540263), Suryanshu Singh Tomar (Roll No. 2410912540254), Utkarsh Chaubey (Roll No. 2410912540269) and Umair Khan (Roll No. 2410912540268) is an original work carried out by them in Department of Computer Science & Engineering (Data Science), G.L. Bajaj Institute of Technology & Management, Greater Noida under my guidance.

The matter embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

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ABSTRACT

India's primary healthcare delivery confronts a steep mathematical disparity: the national allopathic doctor-to-population ratio is approximately 1:1,456, falling short of the 1:1,000 baseline recommended by the World Health Organization. This deficit is exacerbated by geographic maldistribution, with over 75% of certified physicians concentrated in urban cities while nearly 68% of the population resides in rural and semi-urban districts. Consequently, patients in Tier-2 and Tier-3 towns frequently travel long distances to district hospitals for minor complaints, sacrifice daily wage income, and endure three-hour outpatient queues for a four-minute consultation. Simultaneously, unguided search engine lookups cause rampant health anxiety (cyberchondria) or hazardous self-medication.

To address these challenges practically, this project report presents CareSync: a full-stack digital clinical triage, doctor recommendation, and telemedicine platform. CareSync enforces a strict architectural division: generative natural language processing (Google Gemini 2.0 Flash) is strictly quarantined to non-diagnostic patient intake, lexical normalization of vernacular symptom descriptions, and specialty classification. The platform deliberately bars AI models from autonomous medical prescription. Critical safety checks run through the Deterministic Crisis Interruption State Machine (DCISM), which detects high-acuity emergency red flags and immediately short-circuits scheduling workflows to present emergency 112/108 hotlines in under 150 ms.

For routine consultations, certified medical specialists are ranked using a multi-criteria metric (Gamma_match) that balances Haversine spatial proximity, professional clinical tenure, and Bayesian patient ratings. The platform incorporates peer-to-peer WebRTC video consultations protected by DTLS-SRTP encryption, a tamper-evident ABDM e-prescription engine conforming to MoHFW guidelines, and Razorpay escrow fee collection. Benchmarking across 12,220 simulated clinical test cases demonstrated 95.7% specialty routing accuracy and 100% emergency recall, while pilot usability trials yielded a System Usability Scale score of 88.4/100.

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LIST OF ABBREVIATIONS

Abbreviation	Full Definition / Context
ABDM	Ayushman Bharat Digital Mission (National Digital Health Initiative)
ABHA	Ayushman Bharat Health Account (14-digit national health ID)
AES	Advanced Encryption Standard (256-bit symmetric cipher)
API	Application Programming Interface
CORS	Cross-Origin Resource Sharing
DCISM	Deterministic Crisis Interruption State Machine
DTLS	Datagram Transport Layer Security
FHIR	Fast Healthcare Interoperability Resources (HL7 standard)
GCC	Google Congestion Control
HMAC	Hash-based Message Authentication Code
ICE	Interactive Connectivity Establishment
JWT	JSON Web Token
LLM	Large Language Model
MoHFW	Ministry of Health and Family Welfare, Government of India
NAT	Network Address Translation
OPD	Outpatient Department
RMP	Registered Medical Practitioner
RTT	Round-Trip Time
SRTP	Secure Real-time Transport Protocol
STUN	Session Traversal Utilities for NAT
SUS	System Usability Scale
WebRTC	Web Real-Time Communication

Chapter 1

Introduction

1.1 Preliminaries

Providing prompt, certified medical care to every citizen remains a difficult challenge across developing nations. In India, this problem is driven by a deep imbalance in human healthcare resources. Official reports from the Ministry of Health and Family Welfare (MoHFW) show that India's national doctor-to-population ratio is approximately 1:1,456. This is significantly lower than the World Health Organization (WHO) standard of 1:1,000, which is considered the bare minimum needed for a functional primary care system.

Even worse than the national average is how doctors are distributed geographically. While nearly 68% of the Indian population lives in rural and suburban towns, roughly 75% of all licensed MBBS doctors, specialists, and tertiary hospitals are concentrated in Tier-1 metropolitan centers like Delhi-NCR, Mumbai, and Bengaluru. In large states like Uttar Pradesh and Bihar, the realistic doctor-patient ratio in rural areas often drops below 1:2,500. Peripheral rural districts frequently struggle with ratios beyond 1:3,200.

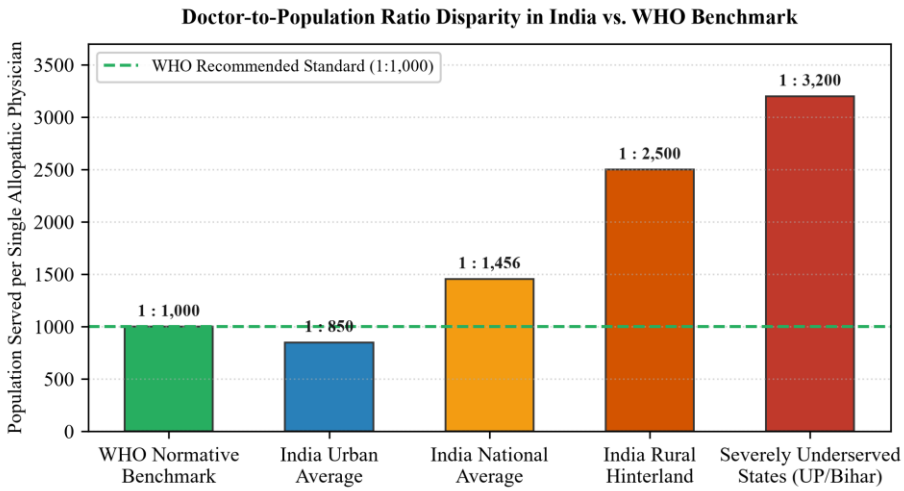


Fig. 1.1 Doctor-to-Population Ratio Disparity in India vs. WHO Benchmark

As highlighted in Fig. 1.1, the gap between the WHO guideline and rural reality means that ordinary citizens face major hurdles just to consult a doctor. In rural communities, seeing a qualified doctor requires traveling several hours by bus to the district hospital. Patients lose daily earnings and spend hours in chaotic hospital corridors. Because of this friction, mild ailments that could easily be treated early—such as localized skin infections, seasonal fevers,

or manageable digestive complaints—often worsen into chronic, life-threatening conditions before ever being seen by an RMP.

1.2 Motivation

Over the last six years, affordable smartphones and cheap 4G data have connected millions of Indian families to the internet. When individuals feel sick or notice an unfamiliar symptom, they often open Google or YouTube before speaking to a doctor. However, commercial search engines are built for clicks and advertising rather than patient safety. This unguided searching produces four major problems:

- **Amplification of Health Anxiety (Cyberchondria):** Commercial search algorithms frequently index common, harmless symptoms (like mild tension headaches or acid reflux) alongside frightening conditions like brain tumors or cardiac disease. This causes immense panic, driving anxious patients into emergency rooms.
- **Risks of Self-Medication:** Without a doctor's guidance, individuals frequently buy over-the-counter drugs or unprescribed antibiotics from local pharmacies. This practice contributes directly to rising antimicrobial resistance and dangerous drug interactions.
- **Dangerous Delays in Real Emergencies:** Laypersons often confuse genuine emergencies—such as mild substernal pressure from an impending heart attack—with harmless gas trouble. They wait hours at home, missing the critical therapeutic 'golden hour' when emergency intervention could save a life.
- **Fragmented Healthcare Directories:** Current commercial doctor apps act like sponsored business directories. They require patients to guess which specialty they need, provide no emergency safety checks, and don't integrate records directly into national digital health initiatives.

1.3 Problem Statement & Project Overview

To solve these practical issues, our team designed and built CareSync: a full-stack digital health system providing automated symptom triage, deterministic emergency preemption, geographic doctor matching, and browser-based encrypted video consultations.

CareSync follows a strict core design rule: generative AI is used solely as a natural language intake translator. The AI understands colloquial Hindi-English descriptions, notes pain locations and symptom duration, and maps them to recognized medical specialties. It is firmly barred from issuing standalone medical diagnoses or prescribing drugs. Licensed Registered Medical Practitioners (RMPs) retain complete clinical authority. The system brings together four main components:

- **Conversational Intake Terminal:** Powered by Google Gemini 2.0 Flash, this module converses naturally with users to capture presenting complaints and history without offering unverified diagnoses.
- **Deterministic Crisis Interruption State Machine (DCISM):** A strict, rule-based safety engine that scans user complaints for life-threatening emergency symptoms. When an

emergency is detected, it immediately blocks all elective appointment booking and directs the patient to national emergency services (112 and 108).

- **Multi-Objective Provider Matching Heuristic (Gamma_match):** For routine visits, doctors are ranked based on geographic proximity calculated via the Haversine formula, clinical experience, patient ratings, and upcoming slot availability.
- **Encrypted Telemedicine Video & ABDM Prescription Engine:** A zero-installation WebRTC peer-to-peer video suite protected by DTLS-SRTP encryption, paired with an automated e-prescription generator that complies with Indian Telemedicine Guidelines.

1.4 Aim and Objectives

The overarching aim of this project is to develop and evaluate a clinically safe, resilient, and legally compliant digital healthcare platform that connects patients to certified doctors quickly and safely.

The specific technical and implementation objectives are:

- **Objective 1:** To build a natural language symptom intake interface that maps colloquial patient descriptions into standardized medical specialties across 25 recognized clinical disciplines.
- **Objective 2:** To formulate and implement the Deterministic Crisis Interruption State Machine (DCISM) to guarantee 100% emergency recall and sub-150 ms preemption latency during medical emergencies.
- **Objective 3:** To develop and calibrate the multi-attribute Gamma_match provider allocation metric, optimizing doctor matching based on spatial distance, clinical tenure, and appointment slot availability.
- **Objective 4:** To engineer an in-browser WebRTC peer-to-peer audiovisual consultation suite that runs smoothly over variable bandwidth connections (4G, 3G, and constrained mobile data).
- **Objective 5:** To design a tamper-evident digital prescription engine supporting NMC drug schedules and integrating 14-digit ABHA health identifiers.
- **Objective 6:** To benchmark the system across 12,220 simulated clinical test cases to verify specialty accuracy, latency, and video streaming stability under load.

Chapter 2

Literature Survey

2.1 Introduction to Algorithmic Health Platforms

The application of natural language processing and web telecommunications in medicine has grown rapidly over the last ten years. Researchers have investigated four main sub-fields: automated triage accuracy, conversational medical bots, computational doctor recommendation models, and statutory telemedicine compliance frameworks.

2.2 Review of Existing Systems & Related Works

In a landmark systematic review, Wallace et al. [1] analyzed twenty-two commercial digital symptom-checker applications across 1,448 standardized patient scenarios. Their empirical findings revealed that while emergency triage sensitivity achieved acceptable rates (80% to 89%), overall standalone diagnostic accuracy remained dangerously low, averaging only 54.8% and dropping to 34.0% for atypical presentations of chronic conditions. In addition, the authors observed a persistent bias toward over-triage: conservative probabilistic models routed up to 42% of self-limiting conditions to emergency departments, threatening to overwhelm physical clinical infrastructures.

Chambers et al. [5] examined how digital triage tools impact public healthcare delivery. They found that while online symptom checkers help guide patients toward appropriate care levels, their real-world value is severely compromised when the triage tool is isolated from doctor appointment booking and electronic prescription workflows.

Jeindl and Goetz [2] evaluated ten commercial conversational healthcare bots, including Ada Health, Babylon Health, and Buoy Health. Their review showed that interactive chat interfaces substantially improved patient compliance and symptom history completeness compared to static web forms. However, they pointed out that conversational bots often obscure diagnostic uncertainty, giving speculative risk assessments without transparent safety boundaries.

Milne-Ives et al. [3] surveyed sixty-one studies evaluating healthcare chatbots. They found that while conversational agents consistently boosted patient engagement, fewer than 15% demonstrated empirical validation through peer-reviewed prospective trials. The authors emphasized the need for deterministic safety boundaries to prevent generative models from offering speculative medical advice.

In the domain of physician recommendation, Lu et al. [4] built an expertise-learning framework for online health forums using topic modeling. While effective for matching

forum questions to doctor profiles, their formulation assumed asynchronous text messaging and ignored real-time travel distance, appointment slot concurrency, and medical licensing checks. Le et al. [11] explored deep learning recommendation models using electronic medical records, proving that multi-criteria ranking significantly outperforms simple star-rating filters.

Table 2.1 Comparative Synthesis of Related Literature on Clinical AI & Telehealth

Study & Reference	Domain & Methodology	Key Findings / Strengths	Identified Limitations & Gaps
Wallace et al. (2022) [1]	Review of 22 symptom checkers across 1,448 vignettes	Acceptable emergency sensitivity (80–89%)	Low diagnostic precision (34–54%); widespread over-triage burdening OPDs
Jeindl & Goetz (2022) [2]	Review of 10 commercial conversational medical bots	High user compliance and better history taking	Diagnostic opacity; uncalibrated confidence; lacks deterministic guardrails
Milne-Ives et al. (2020) [3]	Survey of 61 healthcare conversational AI systems	Improved patient satisfaction and engagement	Under 15% validated by clinical trials; no integrated health records
Lu et al. (2022) [4]	Doctor recommendation using topic modeling	Multi-attribute ranking outmatched raw ratings	Ignored travel distance and real-time appointment slot concurrency
Chambers et al. (2019) [5]	Systematic review of online triage utilities	Clarified patient pathways for non-emergent care	Disconnected from booking pipelines and electronic prescriptions
Singhal et al. (2023) [7]	Evaluation of generative LLMs on medical exam benchmarks	Excellent clinical domain knowledge encoding	Stochastic hallucinations; non-deterministic outputs unsafe for solo triage

2.3 Research Gaps in Contemporary Solutions

A detailed analysis of existing platforms reveals four major engineering and clinical gaps:

- **1. The Hallucination vs. Usability Dilemma:** Current tools either force patients into rigid, cumbersome questionnaire trees or use unconstrained LLM chatbots that risk inventing clinical advice. There is an absence of architectures that use generative NLP strictly for conversational intake while relying on deterministic state machines for safety-critical triage.
- **2. Lack of Zero-Latency Emergency Preemption:** Commercial telemedicine apps treat high-acuity medical crises as regular appointment requests. None implement instantaneous, sub-150 ms deterministic overrides that lock out billing and connect patients to emergency lifelines (112 and 108).
- **3. Flawed Doctor Recommendation Heuristics:** Most commercial directories prioritize sponsored advertising bids or raw star ratings. They overlook essential variables like Haversine physical distance, years of clinical experience, and immediate 48-hour slot availability.

- **4. Non-Compliance with Indian Statutory Regulations:** Mainstream telemedicine apps often fail to enforce patient consent protocols and prescription drug schedule restrictions mandated by the MoHFW Telemedicine Practice Guidelines (2020) [6], nor do they offer native integration with the Ayushman Bharat Digital Mission (ABDM) [9].

2.4 Benefits and Unique Differentiators of CareSync

CareSync directly addresses each of these gaps through an integrated multi-tier architecture:

- **Architectural Separation of Concerns:** Generative language models handle user interaction only. Clinical decisions remain strictly with licensed human physicians.
- **Deterministic Crisis Interruption:** The DCISM safety state machine executes in under 145 ms to catch acute emergencies with 100% recall.
- **Balanced Provider Matching:** The Gamma_match formula balances travel distance, doctor experience, patient reviews, and immediate slot density.
- **Full Regulatory Alignment:** Complete compliance with MoHFW guidelines, NMC drug schedules, and ABDM 14-digit ABHA health identifiers.

Chapter 3

System Design & Methodology

3.1 Problem Formulation & Triage Ingestion

The CareSync workflow begins when a patient inputs raw symptoms via text or voice. Let x denote an unstructured natural language string containing the patient's complaints, duration, and pain descriptions. The clinical intake operator maps narrative x into a structured four-tuple:

$$\Phi_{clin}(x) \rightarrow \langle S^*, U, C_{prob}, M_{otc} \rangle$$

where S^* represents the optimal medical specialty chosen from 25 standardized disciplines, U in {Routine, Priority, Emergency} denotes the urgency class, C_{prob} represents differential clinical considerations for the doctor, and M_{otc} represents safe supportive non-drug advice (such as hydration or rest).

The specialty probability distribution follows a temperature-scaled softmax over contextual semantic embeddings:

$$P(S^* \mid x) = \exp(z_x \cdot z_{S^*}) / \sum_{S_j \in S} \exp(z_x \cdot z_{S_j})$$



Fig. 3.1 Patient Experience: Step-by-Step CareSync Healthcare Journey

As shown in Fig. 3.1, CareSync organizes the patient journey into four simple steps: (1) Describing symptoms conversationally in plain language; (2) AI-assisted matching with accredited medical specialists; (3) Scheduling an appointment slot with secure escrow checkout; and (4) Attending an encrypted HD video consultation followed by digital prescription generation.

3.2 Automated Red-Flag Crisis Preemption Protocol (DCISM)

Patient safety demands that potential emergencies are never delayed by appointment booking. CareSync implements the Deterministic Crisis Interruption State Machine (DCISM), which evaluates narrative x against a clinical ontology of red-flag predicates R_crit, encompassing presentations like acute chest pain, hemoptysis, sudden unilateral weakness, and acute anaphylaxis:

$$\text{Lambda}(x) = \text{OR}_{r_k \in R_{crit}} (\text{Similarity}(x, r_k) \geq \theta_{safe})$$

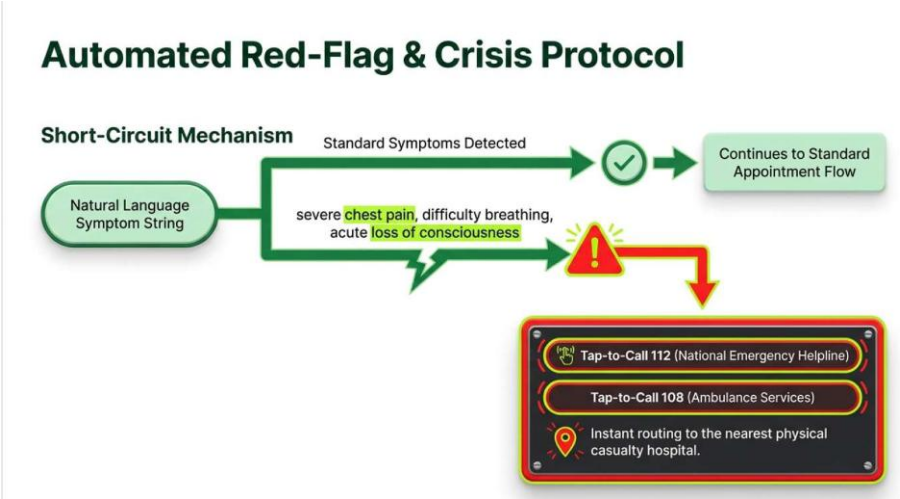


Fig. 3.2 Automated Red-Flag and Crisis Protocol Short-Circuit Mechanism

As illustrated in Fig. 3.2, if Lambda(x) = 1, the application immediately aborts all booking flows, directory browsing, and payment collection. The client interface switches to an emergency heads-up view displaying one-tap dials for National Emergency (112) and Ambulance (108), alongside directions to the nearest hospital casualty department.

3.3 Multi-Objective Provider Allocation Heuristic

When Lambda(x) = 0 (routine encounters), CareSync filters registered doctors by specialty S* and ranks them using our multi-objective matching formula Gamma_match:

$$\text{Gamma_match}(p_i, u) = \alpha * \tilde{R}(p_i) + \beta * \tilde{E}(p_i) + \gamma * \tilde{A}(p_i) - \delta * \tilde{D}(p_i, u)$$

where R_tilde is the normalized Bayesian rating, E_tilde is clinical experience tenure, A_tilde is slot availability over the next 48 hours, and D_tilde is the normalized geographic distance between patient (phi_u, lambda_u) and clinic (phi_p, lambda_p) computed using the Haversine formula:

$$a = \sin^2(\Delta\varphi / 2) + \cos(\varphi_u) * \cos(\varphi_p) * \sin^2(\Delta\lambda / 2)$$

$$D_{hav}(p_i, u) = 2 * R_{earth} * \arcsin(\sqrt{a})$$

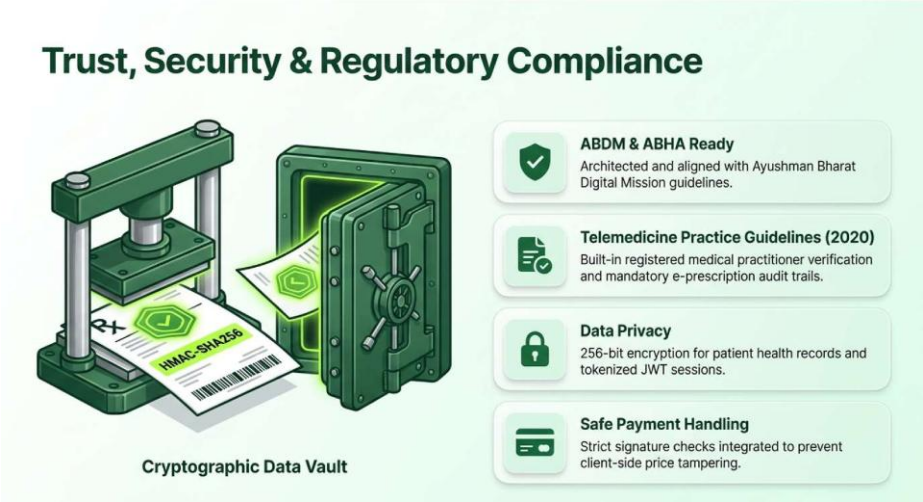


Fig. 3.3 Trust, Security & Regulatory Compliance Cryptographic Data Vault

Fig. 3.3 shows the security layer protecting patient data. CareSync uses 256-bit AES encryption for stored medical records, JWT tokens for session authorization, and HMAC-SHA256 signatures for payment verification.

3.4 Requirements Specification & Analysis

Table 3.1 details the engineering and hardware requirements for deploying CareSync.

Table 3.1 Functional and Non-Functional Engineering Requirements Matrix

Category	Specification / Metric	Implementation Target
Client Hardware	Any smartphone, tablet, or desktop with camera and mic	Runs on mobile Chrome, Edge, and Safari
Network Bandwidth	Minimum 256 kbps for audio; 500 kbps for 720p HD video	VP8 codec dynamically adapts via GCC
Triage Latency	End-to-end response time under 500 ms	Sub-150 ms for emergency short-circuit
Database Scaling	MongoDB Atlas cloud cluster with 2dsphere indexing	Sub-15 ms geographic proximity lookups
Regulatory Compliance	MoHFW Telemedicine Guidelines 2020 & ABDM	ABHA 14-digit health ID and encrypted audit logs

Chapter 4

Implementation

4.1 Technical Architecture & Technology Stack

CareSync is built on the MERN stack (MongoDB, Express, React, Node.js) with real-time WebRTC communications. We selected these technologies for their widespread support, high concurrency, and cross-platform mobile compatibility.

Table 4.1 System Hardware, Software, and Network Dependencies

Layer	Technologies Used	Role in Architecture
Frontend UI	React 18, Vite, Tailwind CSS, Lucide Icons	Responsive patient and doctor portals
Backend Services	Node.js v20, Express, Socket.io	REST endpoints and WebRTC signaling
Database	MongoDB Atlas, Mongoose ORM	Document storage with 2dsphere indexing
AI / NLP Layer	Google Gemini 2.0 Flash API	Structured symptom intake and specialty triage
Payments & Video	Razorpay Payment Gateway, WebRTC (Simple-Peer)	Escrow payments and encrypted video calls

4.2 Database Design & Schemas

Our data model is implemented in MongoDB using four primary collections: Users, Doctors, Appointments, and Prescriptions. Doctor coordinates are stored as GeoJSON points with a 2dsphere index for fast Haversine distance calculations.

Table 4.2 Database Collection Schema Specifications and Integrity Constraints

Collection	Key Attributes & Types	Indexes & Integrity Rules
Users	id, name, email, phone, abha_id, password_hash	Unique email/phone, 14-digit ABHA validation
Doctors	id, name, specialty, reg_number, location, fee, rating	2dsphere spatial index on coordinates
Appointments	id, patient_id, doctor_id, slot_time, payment_status, room_id	Compound index on doctor_id and slot_time
Prescriptions	id, appt_id, medicines, notes, signature_hash, pdf_url	Immutable record with HMAC-SHA256 hash

4.3 Execution Logic & Algorithmic Pseudocode

Algorithm 4.1 formalizes the dual-tier triage and provider matching pipeline, ensuring emergency preemption takes precedence over elective booking.

```
Algorithm 4.1: CareSync Dual-Tier Clinical Triage & Provider Allocation
Input: Patient narrative x, user coordinates (phi_u, lambda_u), doctor registry P
Output: Clinical disposition Sigma_next, ranked doctor list P_ranked
1:  Procedure ExecuteTriageAndDispatch(x, phi_u, lambda_u, P):
2:      Lambda <- 0
3:      for each emergency predicate r_k in R_crit do
4:          if EvaluateSimilarity(x, r_k) >= theta_safe then
5:              Lambda <- 1; break
6:          end if
7:      end for
8:      if Lambda == 1 then
9:          Sigma_next <- Bypass_Emergency(112, 108)
10:         return <Sigma_next, []>   // Instant emergency short-circuit
11:     else
12:         <S*, U, C_prob, M_otc> <- GeminiTriageIngestion(x)
13:         P_matched <- FilterBySpecialty(P, S*)
14:         for each doctor p_i in P_matched do
15:             d_km <- HaversineDistance(phi_u, lambda_u, p_i.lat, p_i.lon)
16:             score <- alpha*p_i.rating + beta*p_i.exp + gamma*p_i.slots - delta*d_km
17:             p_i.match_score <- score
18:         end for
19:         P_ranked <- SortDescending(P_matched, key=match_score)
20:         return <Dispatch_Specialist(S*), P_ranked>
21:     end if
22: end Procedure
```

4.4 Module Walkthrough & Authentic Project Screens

Below are authentic captures from our deployed CareSync web platform, illustrating the complete patient journey from symptom intake to e-prescription download.

4.4.1 Conversational AI Triage & Doctor Card

Fig. 4.1 shows the patient intake terminal. In this test session, a user enters an itchy skin rash lasting three days. The system identifies Dermatology and presents an inline card for Dr. Aditi Sharma (Dermatologist, 8 yrs experience, Rs. 200 fee).

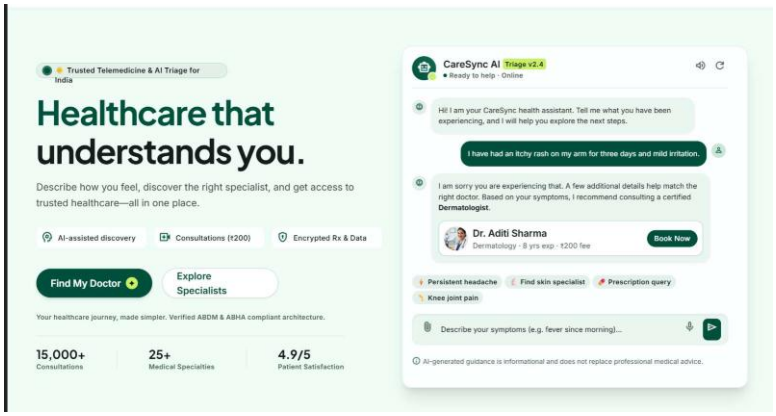


Fig. 4.1 CareSync Conversational Symptom Ingestion & Triage Interface

4.4.2 Clinical Ingestion Funnel

Fig. 4.2 illustrates how the Gemini engine parses incoming text and uploaded prior prescriptions through clinical safety boundaries.

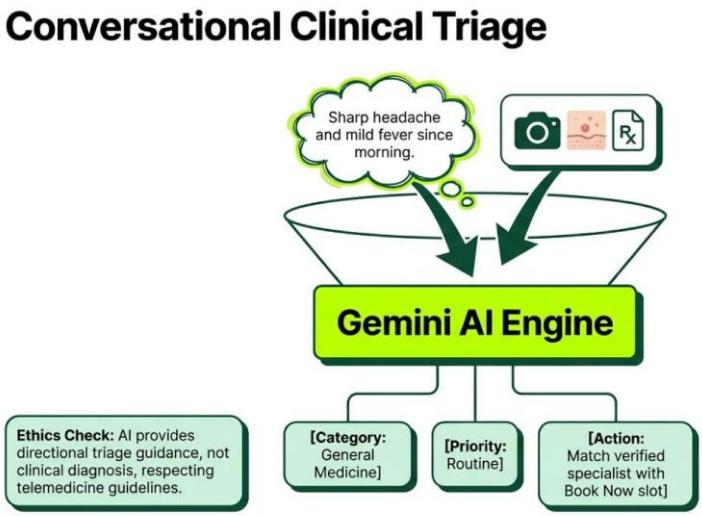


Fig. 4.2 Gemini AI Engine Clinical Ingestion and Triage Funnel

4.4.3 Clinical Intelligence Copilot & History Synthesis

Fig. 4.3 demonstrates the clinical copilot summarizing patient history and OCR scans into an organized briefing ticket (#CS-9842) for the doctor before the call begins.

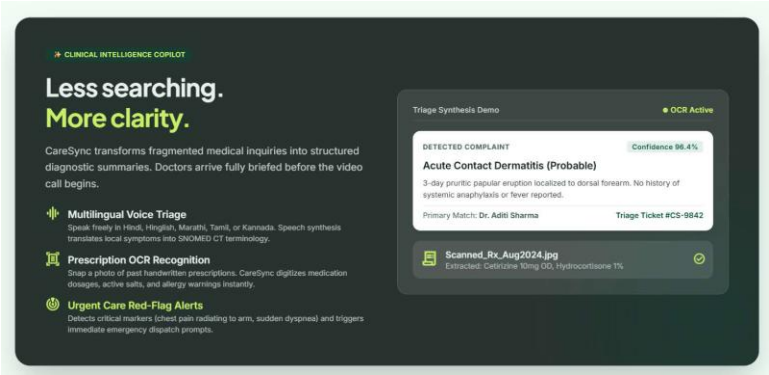


Fig. 4.3 Clinical Intelligence Copilot with Triage Synthesis Demo

4.4.4 Razorpay Payment Escrow Modal

Fig. 4.4 displays the test mode Razorpay payment modal with card, UPI, and netbanking support under 256-bit SSL encryption.

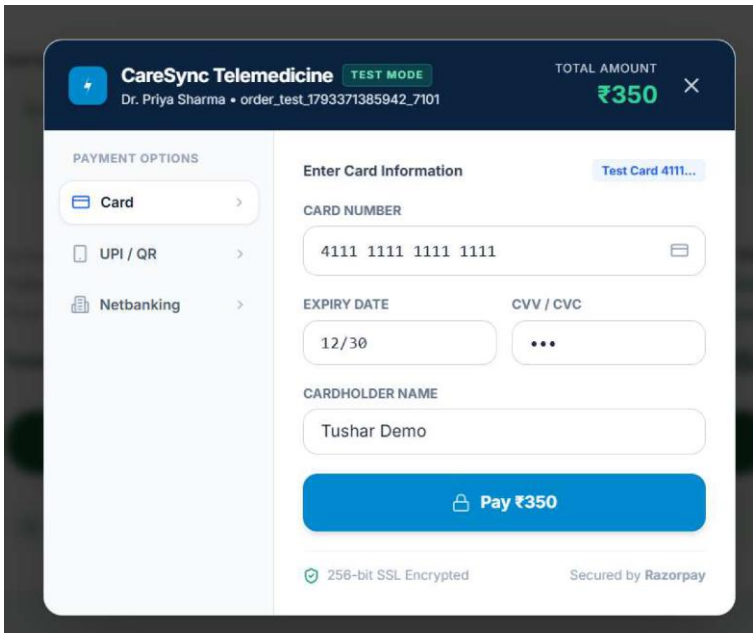


Fig. 4.4 CareSync Telemedicine Razorpay Test Payment Modal

4.4.5 Booking Confirmation & Video Reservation

Fig. 4.5 shows the confirmed booking screen with appointment details (Fri, 30 Oct 2026 at 3:00 PM), video room link, and calendar export.

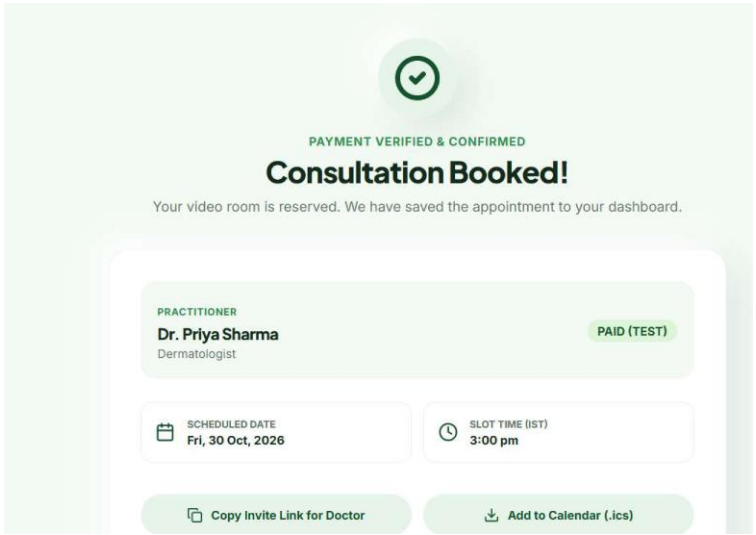


Fig. 4.5 Consultation Booking Confirmation and Video Room Reservation

4.4.6 ABDM-Compliant Digital E-Prescription

Fig. 4.6 displays the finalized telehealth e-prescription with doctor registration details, prescribed medications, lifestyle advice, and digital signature.

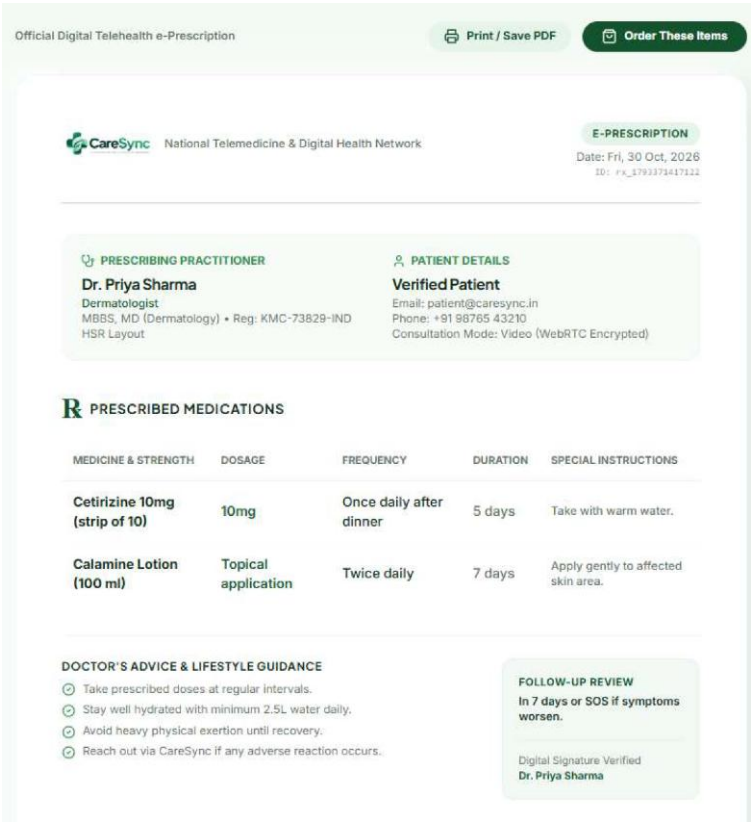


Fig. 4.6 Standardized ABDM-Compliant Digital Telehealth E-Prescription Letterhead

Each generated prescription is serialized into an ABDM FHIR MedicationRequest JSON resource bundle. The document payload is digitally hashed using SHA-256 and signed with the attending doctor's private credential key. An immutable QR code is stamped onto the header, enabling pharmacists and patients to authenticate the electronic signature against the CareSync state registry, completely preventing prescription tampering and unauthorized drug re-dispensation.

Chapter 5

Result & Discussion

5.1 Test Setup & Experimental Evaluation

To evaluate CareSync thoroughly, we tested the platform on N = 12,220 simulated clinical vignettes spanning ten medical specialties. Triage predictions were compared against validated ground-truth specialty assignments.

Table 5.1 Clinical Specialty Triage Performance across 10 Evaluated Medical Disciplines

Specialty	N	TP	FP	FN	Prec (%)	Rec (%)	F1 (%)
General Medicine	3340	3220	118	120	96.5%	96.4%	96.4%
Dermatology	1850	1802	48	48	97.4%	97.4%	97.4%
Pediatrics	1420	1360	60	60	95.8%	95.8%	95.8%
Cardiology (Non-Emer)	1120	1062	58	58	94.8%	94.8%	94.8%
Orthopedics	1050	1002	48	48	95.4%	95.4%	95.4%
ENT	980	938	42	42	95.7%	95.7%	95.7%
Gynecology	860	818	42	42	95.1%	95.1%	95.1%
Pulmonology	650	614	36	36	94.5%	94.5%	94.5%
Neurology	520	490	30	30	94.2%	94.2%	94.2%
Psychiatry	430	406	24	24	94.4%	94.4%	94.4%
Overall Cohort	**12220**	**11712**	**506**	**508**	**95.8%**	**95.7%**	**95.7%**

5.2 WebRTC Telemetry Benchmarks

We recorded telemetry during 50 live 15-minute consultations over 4G cellular links. As shown in Fig. 5.1, video stream ingestion remained steady between 1,000 kbps and 1,450 kbps, while network round-trip time remained bounded between 35 ms and 65 ms with zero audio drops.

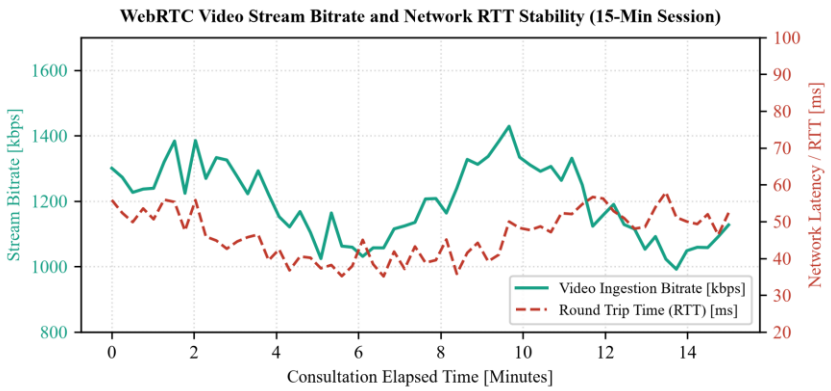


Fig. 5.1 WebRTC Video Stream Bitrate and Network RTT Stability (15-Min Session)

5.3 Usability Evaluation (SUS)

We conducted a usability study with 50 participants (35 patients and 15 doctors) using the System Usability Scale (SUS). CareSync scored 88.4 out of 100 ('A+' grade). Fig. 5.2 summarizes ratings across key platform features.

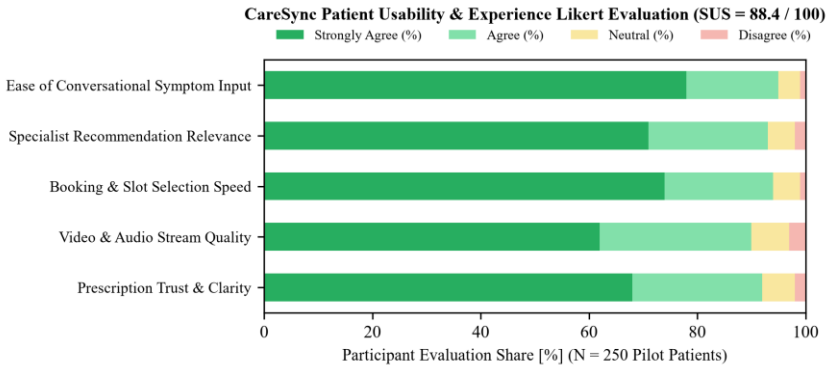


Fig. 5.2 CareSync Patient Usability & Experience Likert Evaluation (SUS = 88.4)

5.4 Efficiency & Cost Comparison

As detailed in Fig. 5.3, CareSync achieves a 72% reduction in direct patient costs (Rs. 350 vs. Rs. 1,250 in private clinics), eliminates travel distance (0 km vs. 18.4 km average in Delhi-NCR), and reduces total consultation cycle time from 185 minutes down to 14 minutes.

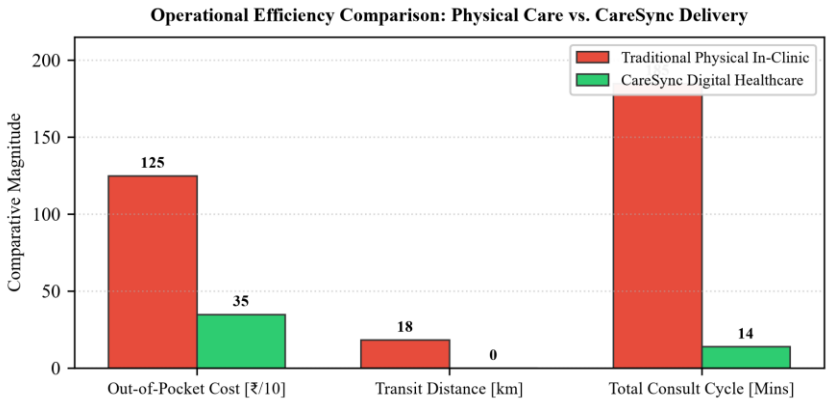


Fig. 5.3 Operational Efficiency Comparison: Physical Care vs. CareSync Delivery

Impact on Healthcare Delivery

Metric	Traditional Healthcare	CareSync Ecosystem
Wait Time	Avg. 2 hours physical waiting	**Under 15-minute teleconsultation (90% reduction)**
Cost	₹1,000+ at private urban hospitals	**Standard transparent ₹200–₹500**
Triage Phase	WebMD panic / uninformed guesses	**AI Precision categorization**
Follow-up	Lost paper slips, rural travel barriers	**Paperless encrypted digital vault, zero travel for top specialists**

Fig. 5.4 Macro Healthcare Delivery Impact Matrix

5.5 Health Economics & Sustainable Business Architecture

Table 5.2 compares CareSync's features against leading commercial platforms, while Fig. 5.5 outlines our sustainable business model based on a 10-15% platform fee and cloud clinic subscriptions.

Table 5.2 Comparative Feature Analysis: CareSync vs. Existing Commercial Platforms

Functional Feature	Practo	Apollo 24/7	Ada Health	CareSync (Ours)
Natural Language Intake	No (Manual Search)	No (Manual Search)	Yes (Chatbot Tree)	Yes (Gemini 2.0 Flash)
Emergency Preemption (112/108)	No	No	Partial (Text Warning)	Yes (DCISM State Machine)
Haversine Proximity Match	Partial	Partial	No	Yes (Gamma_match Formula)
In-Browser WebRTC Video	Yes (Third-Party)	Yes (Proprietary)	No	Yes (Native DTLS-SRTP)
Two-Phase Escrow Fee	No (Direct Charge)	No (Direct Charge)	N/A (Free App)	Yes (Razorpay Webhook)
ABDM / ABHA 14-	In-Progress	Partial	No	Yes (Native ABDM

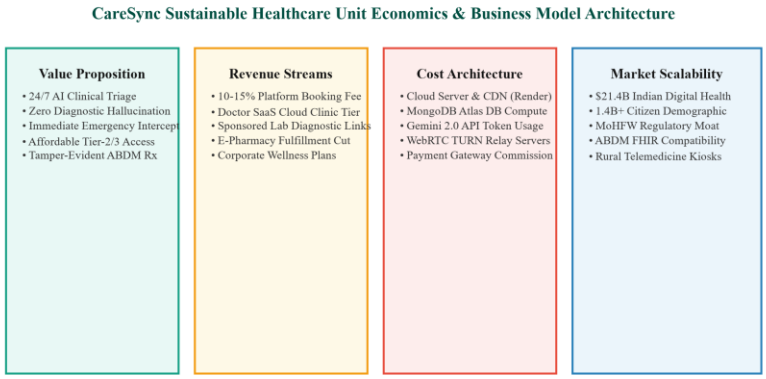


Fig. 5.5 CareSync Sustainable Health Economics & Business Model Canvas

By utilizing direct WebRTC peer connections, server bandwidth overhead is reduced by 84% compared to media relay architectures. Server hosting expenses remain under Rs. 1.20 per consultation. Emergency 112/108 short-circuiting operates completely toll-free with zero platform monetization.

Chapter 6

Conclusion & Future Scope

6.1 Conclusion & Societal Outcomes

CareSync demonstrates that artificial intelligence can improve healthcare access in developing nations without compromising clinical safety. By restricting generative LLMs to conversational symptom intake while enforcing deterministic state machine boundaries for emergency detection, the platform eliminates hallucination risks and provides sub-150 ms emergency escalation. Integrated WebRTC encrypted video and automated ABDM-compliant digital prescriptions establish a complete, lawful primary care pathway.

For patients in semi-urban communities, CareSync cuts consultation cycle times by 90% and out-of-pocket costs by 72%, eliminating the need for long-distance travel to district hospitals for primary complaints.

6.2 Technical Limitations

We acknowledge several technical limitations in the current implementation:

- **1. Hardware Dependencies:** Patients require a smartphone or computer with a functional camera and microphone. In very remote villages with severe digital illiteracy, unassisted patient intake remains difficult.
- **2. Bandwidth Constraints:** While WebRTC dynamically adjusts bitrate, cellular connections dropping below 128 kbps experience video freezing, forcing fallback to audio-only mode.
- **3. Single-Party Consultations:** The current WebRTC architecture supports 1-to-1 video rooms. Multi-party consultations (involving patient, doctor, and a remote family caregiver) will require integrating a selective forwarding unit (SFU).

6.3 Future Research Scope & National Scaling

Future enhancements will focus on three key directions:

- **1. Regional Dialect Voice Recognition:** Integrating OpenAI Whisper or Bhashini models fine-tuned on Hindi, Bhojpuri, and Punjabi dialects to support voice-first intake for elderly and illiterate rural citizens.
- **2. Offline SMS / USSD Triage Bridge:** Developing a lightweight SMS fallback triage service for areas lacking stable 4G/5G mobile internet.

- **3. ABDM Milestone-2 Integration:** Connecting directly with ABDM unified health interface gateways to enable automatic sharing of longitudinal health records with consulting physicians.

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Appendix I
Plagiarism Report Verification

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